

Artificial Intelligence (AI) Patent Production by Geographic Region

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MAE298 – STATISTICS AND DESIGN OF EXPERIMENTS

Overview

- Introduction
- Datasets
- Treatment of the Data
- Exploratory Plots
- Modeling
- Conclusions
- Questions?

Introduction

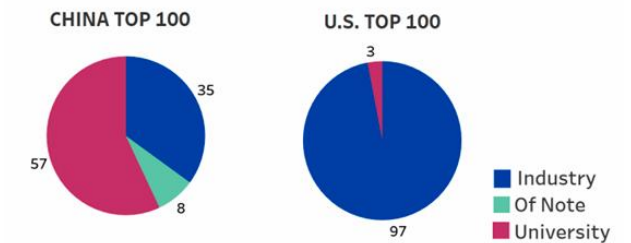
AI is a transformative tool that is seeing a surge of investment and interest from global superpowers

- Genesis Mission (USA)

AI technology dominance is shaping current economic progress

- Machine learning
- Computer vision
- Robotics
- Generative AI

What factors influence AI patent production across regions over time?



Source: Center for Security and Emerging Technology

Stanford AI Index Report 2025

Main dataset used for analysis (Eighth edition)

- 1. Research and Development – patents, publications, citations, # of notable models
- 2. Technical Performance
- 3. Responsible AI
- 4. Economy – private investment
- 5. Science and Medicine
- 6. Policy
- 7. Education
- 8. Public Opinion

World Bank World Development Indicators (WDI)

- Economic data – GDP, R&D expenditure, education expenditure

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Regions

These are grouped regions from the Stanford AI Index

Asia ~ China, S. Korea, and Japan

Europe ~ UK, Germany, France

India

US

Number of notable AI models by geographic area, 2003–24 (sum)

Source: Epoch AI, 2025 | Chart: 2025 AI Index report

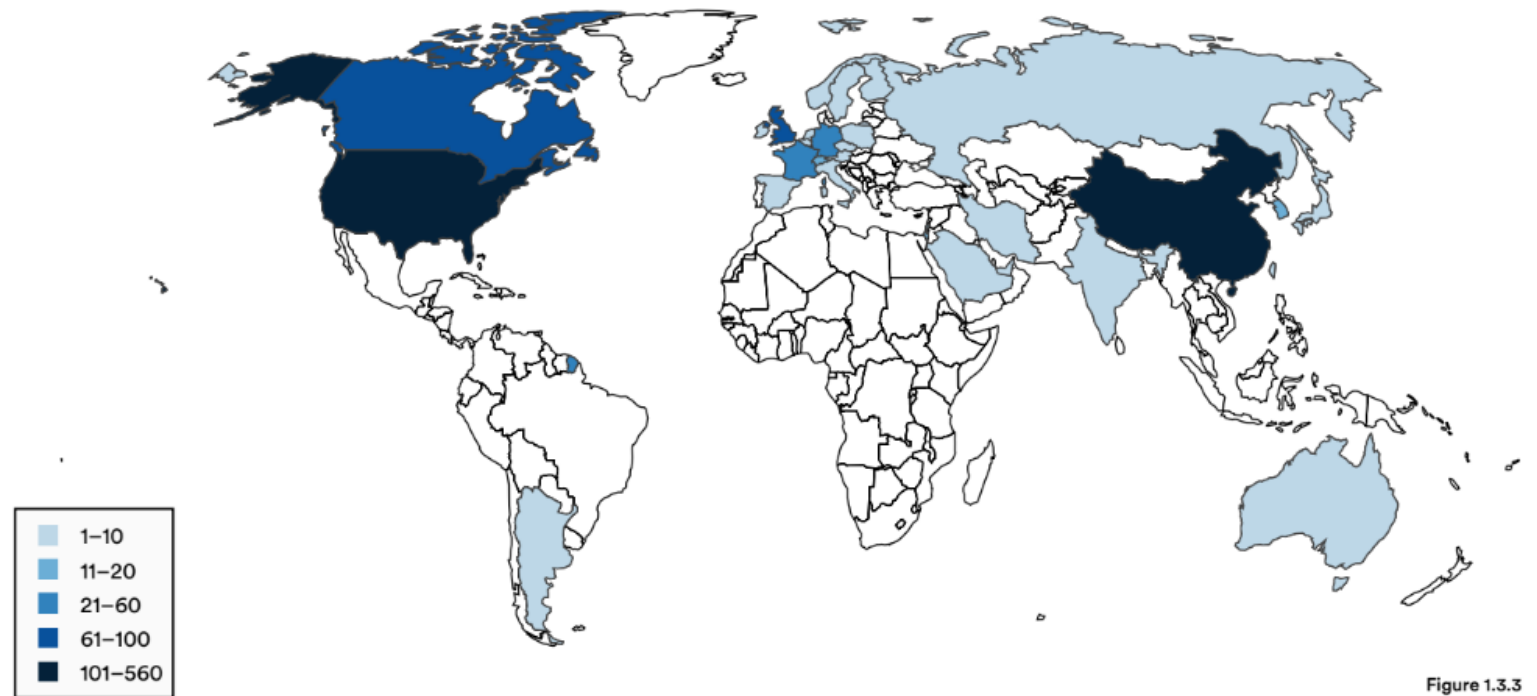


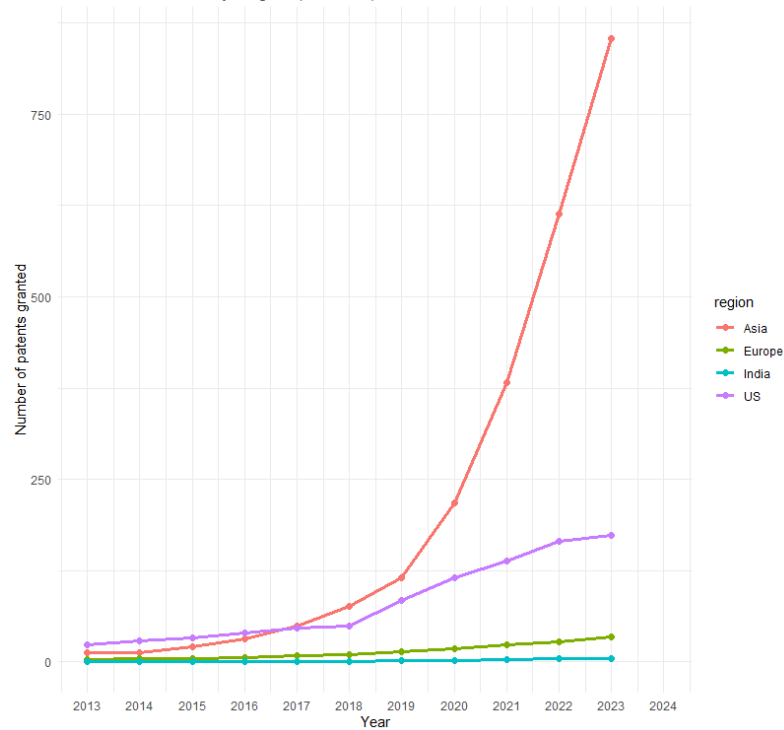
Figure 1.3.3

Data Panel – Longitudinal (2013-2023)

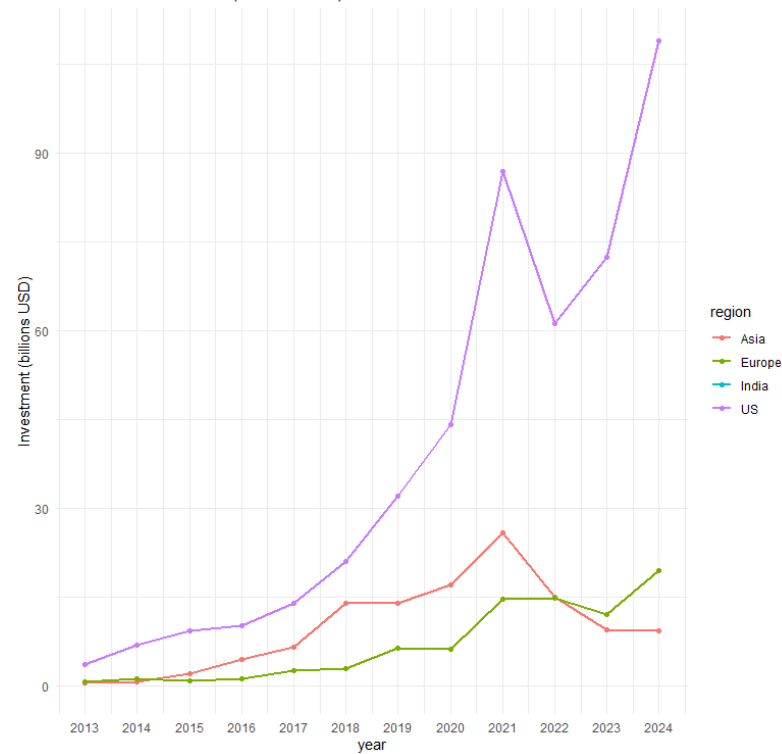
Year	Integer
Region	Categorical
Publications (% of global total)	Numeric
Publications (global total)	Numeric
Absolute Publications (per region)	Numeric
Citations (% of global total)	Numeric
Patents (% of global total)	Numeric
Investment (USD)	Numeric
GDP (per region)	Numeric
R&D GDP Weighted	Numeric
R&D Expenditure (USD)	Numeric

Exploratory Plots

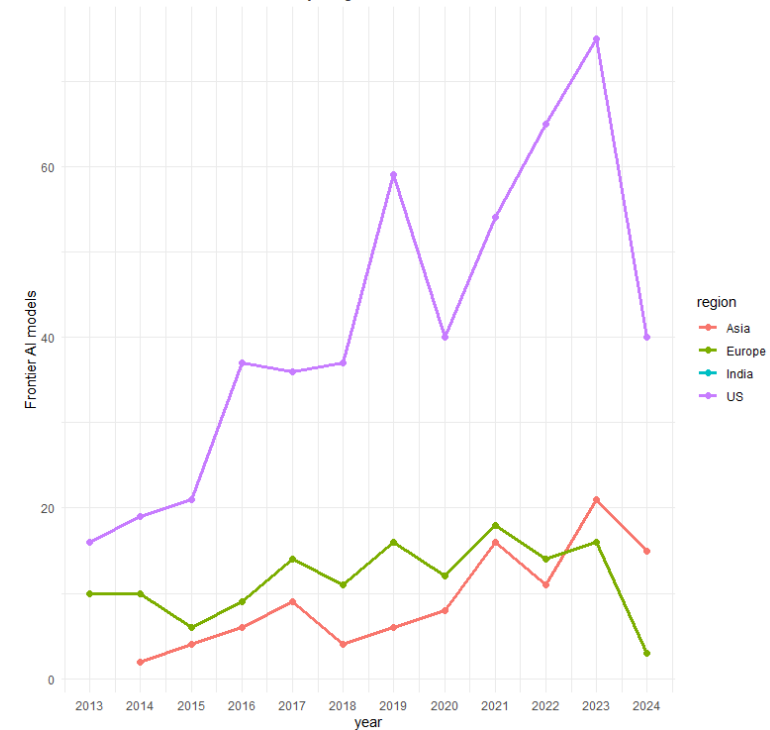
Granted AI Patents by Region (Absolute)



Private AI Investment (USD billions)

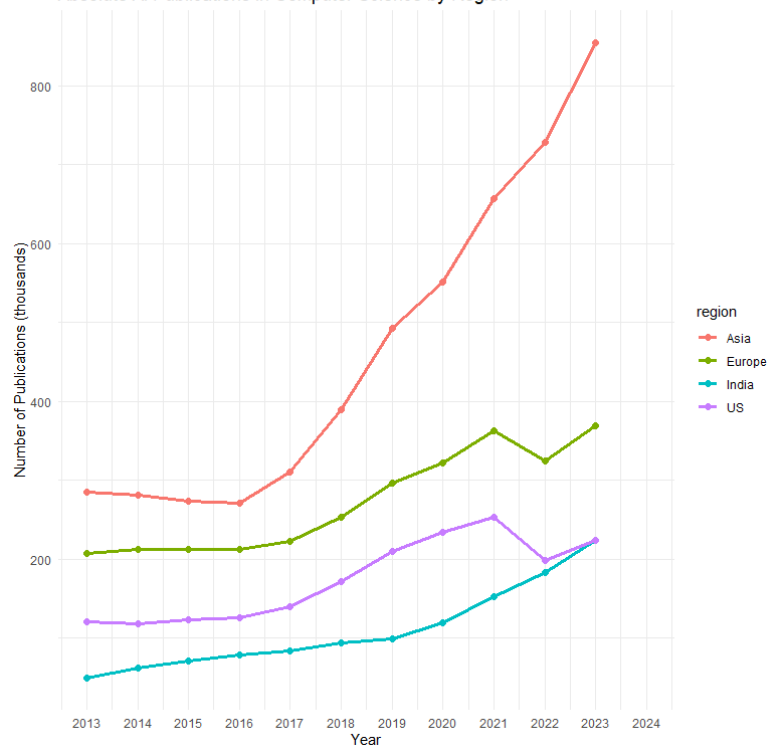


Number of Notable AI Models by Region

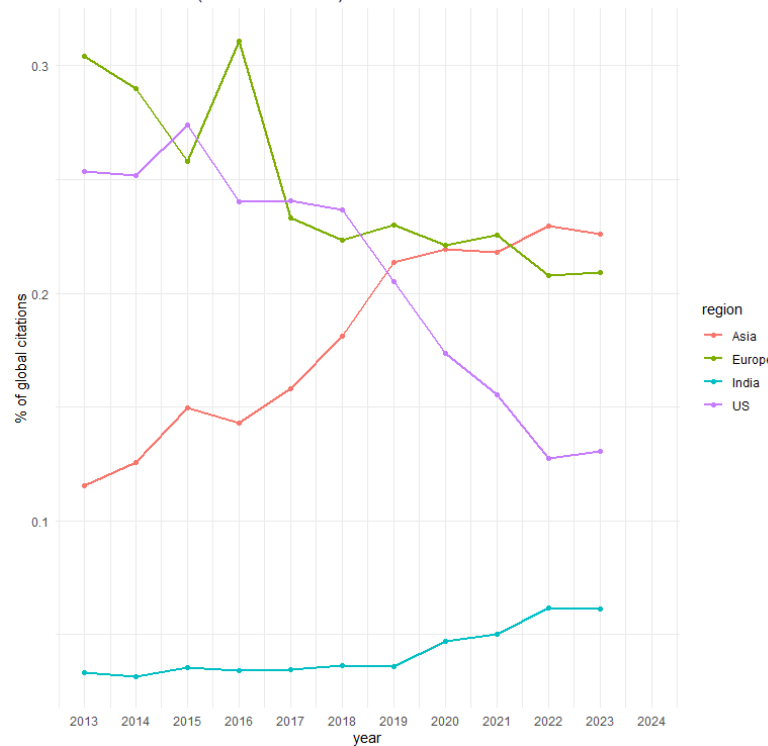


Exploratory Plots

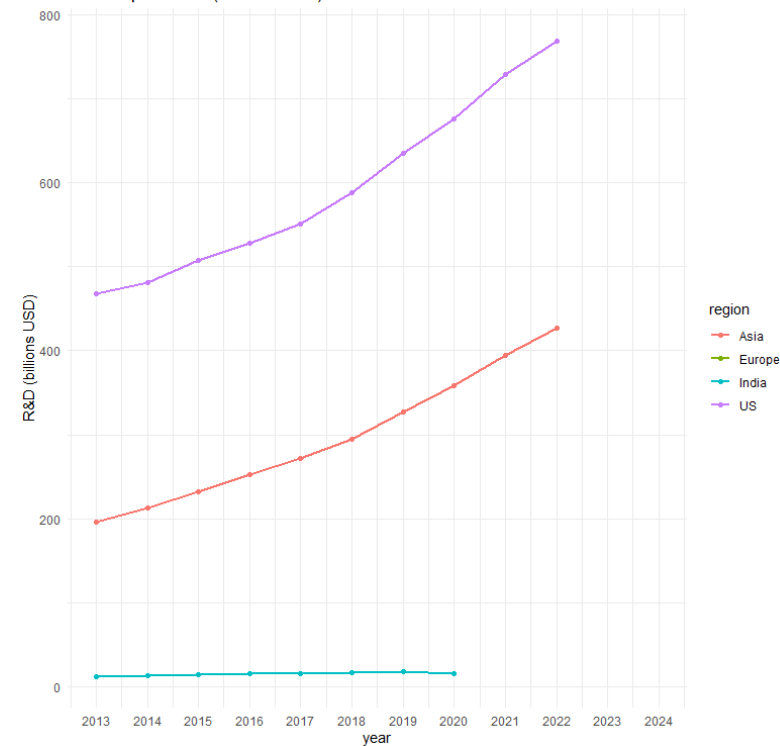
Absolute AI Publications in Computer Science by Region



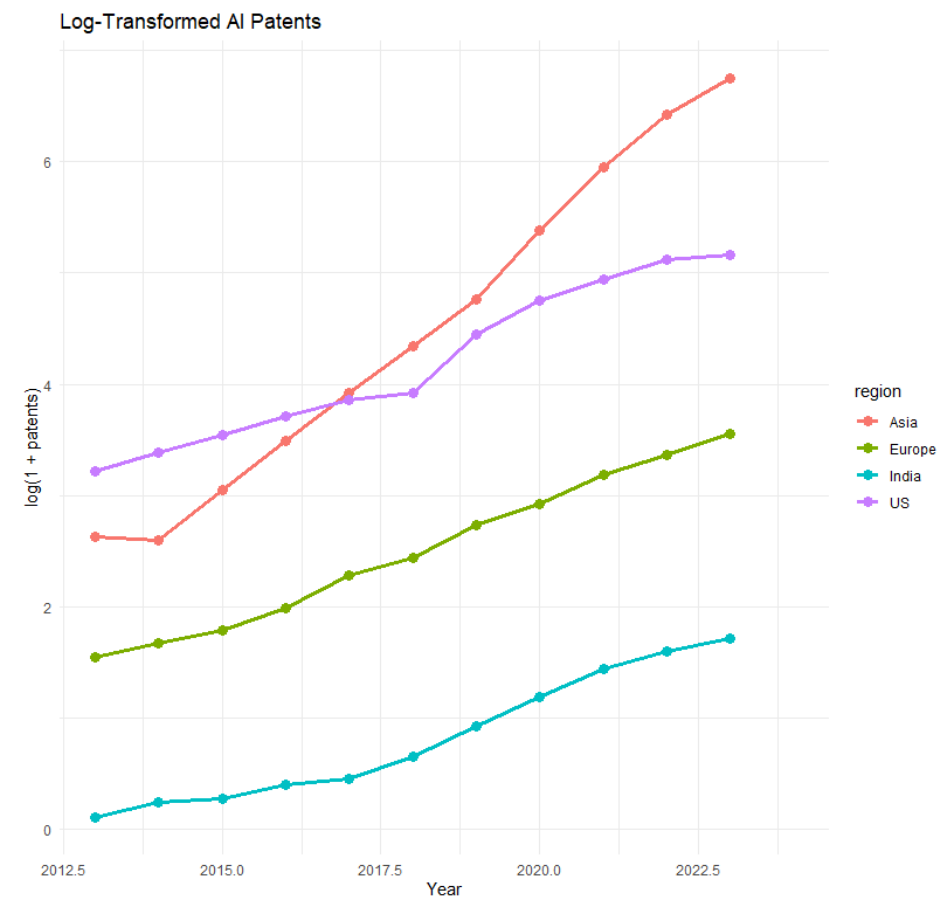
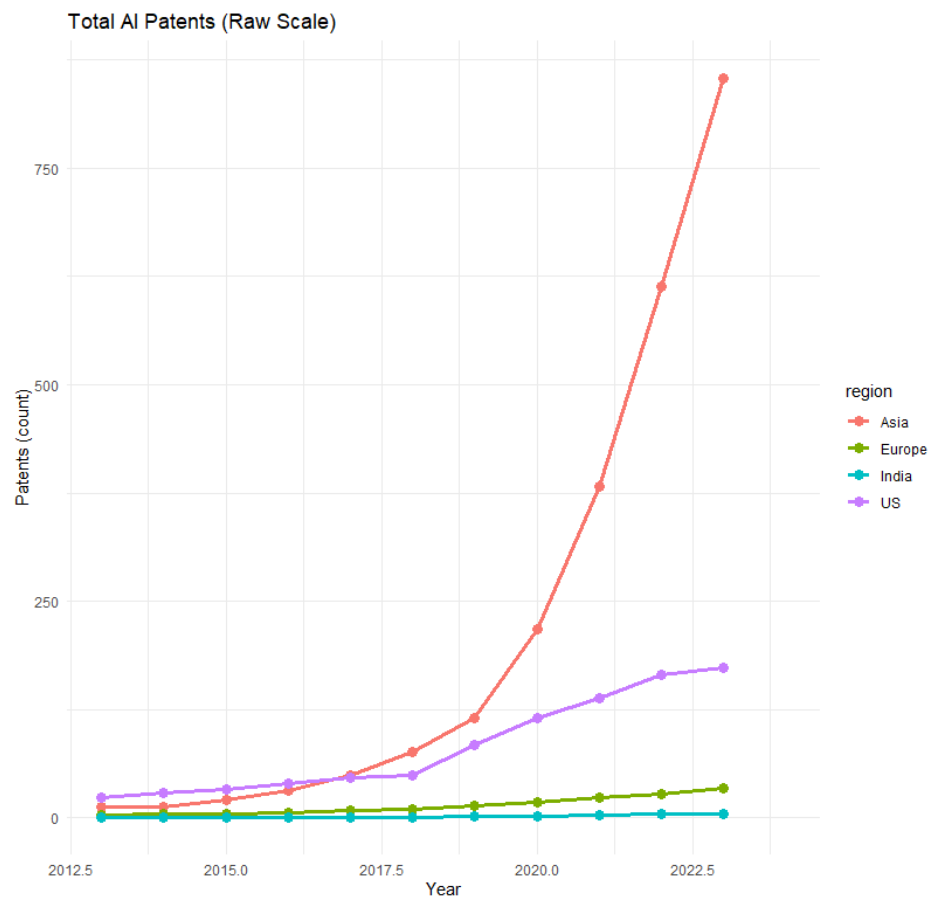
AI Citation Share (% of World Total)



R&D Expenditure (USD billions)



Log Transform

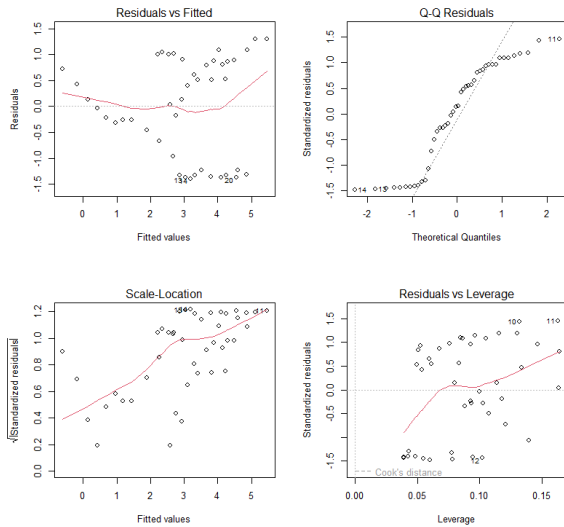


Ordinary Least Squares Model

Dependent variable: issued patents

Independent variables: total publications, GDP, year

Data is log transformed (due to right skew and exponential growth)



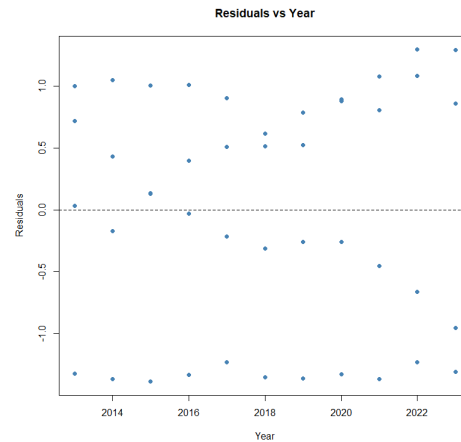
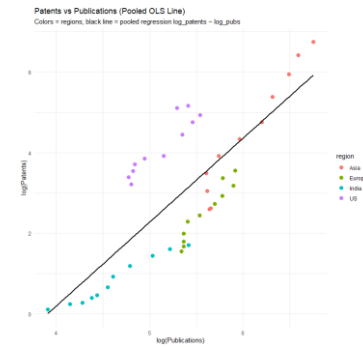
```
> summary(m_ols)

call:
lm(formula = log_patents ~ log_pubs + log_gdp + year, data = df_ols)

Residuals:
    Min       1Q   Median       3Q      Max
-1.3916 -1.0273  0.1320  0.8643  1.2962

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -282.54757   113.95964   -2.479  0.017474 *
log_pubs      1.00641    0.35665    2.822  0.007401 **
log_gdp       0.85638    0.22364    3.829  0.000443 ***
year          0.12609    0.05598    2.253  0.029838 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.97 on 40 degrees of freedom
Multiple R-squared:  0.712,    Adjusted R-squared:  0.6904
F-statistic: 32.96 on 3 and 40 DF,  p-value: 6.764e-11
```



Fixed Effects Model

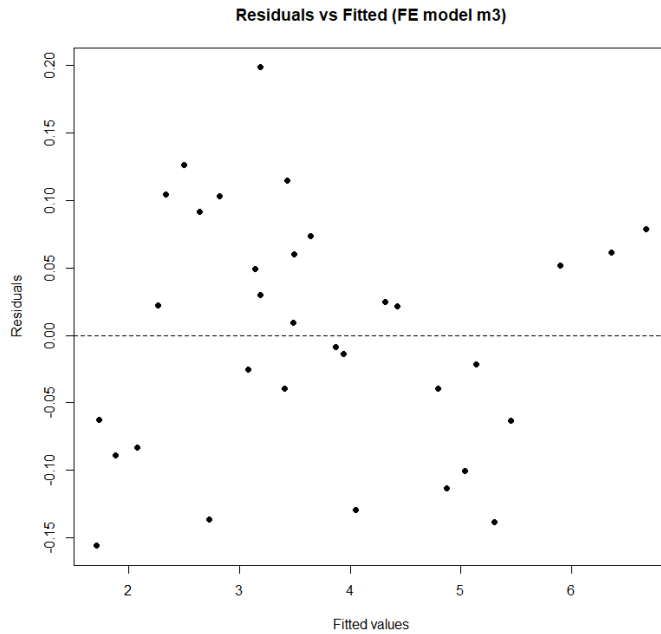
Dependent variable: issued patents

Independent variables: publications, GDP, private investment, R&D spending

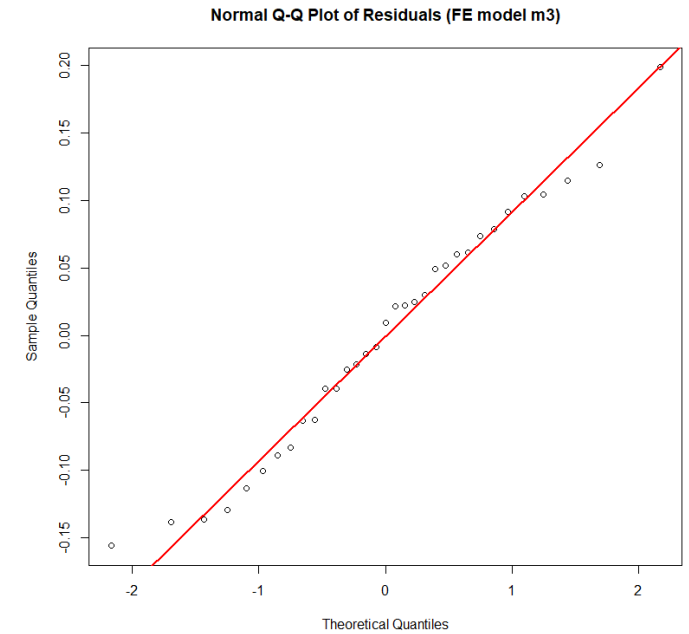
	m1	m2	m3	m4
Dependent var.:	log_patents	log_patents	log_patents	log_patents
log_pubs	0.8078 (0.5192)	-0.6070 (0.6944)	0.9777* (0.4454)	1.033** (0.3932)
log_gdp		3.921* (1.421)	4.624*** (0.7056)	1.11* (2.984)
log_inv			0.0081 (0.0969)	0.2996* (0.1059)
log_rd				-0.189. (3.889)
Fixed-Effects:	-----	-----	-----	-----
region	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
S.E. type	IID	IID	IID	IID
AIC	58.994	50.404	-35.024	-44.704
BIC	85.757	78.951	-11.079	-29.768
R2	0.96189	0.97004	0.99560	0.99866
within R2	0.07705	0.27448	0.93982	0.98732

	log_rd	log_gdp	log_inv	log_pubs
log_rd	1.0000000	0.9887233	0.82473029	-0.40478303
log_gdp	0.9887233	1.0000000	0.83882467	-0.37833224
log_inv	0.8247303	0.8388247	1.00000000	0.07485766
log_pubs	-0.4047830	-0.3783322	0.07485766	1.00000000

$$\log(\text{patents})_{it} = \beta_1 \log(\text{pubs})_{it} + \beta_2 \log(\text{gdp})_{it} + \beta_3 \log(\text{investment})_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$



```
OLS estimation, Dep. var.: log_patents
Observations: 33
Fixed-effects: region: 3, year: 11
Standard-errors: IID
      Estimate Std. Error t value Pr(>|t|)
log_pubs 0.977718   0.445372  2.195283 4.2315e-02 *
log_gdp  4.623656   0.705627  6.552548 4.9270e-06 ***
log_inv  0.008114   0.096920  0.083715 9.3426e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.087646   Adj. R2: 0.99172
                  within R2: 0.939819
```



Fixed Effects Model

Conclusions

Model explains 94% of the variation over time

A 1% increase in AI publications is associated with ~0.98% more AI patents within the same region, controlling for GDP and investment

GDP is the dominant driver, a 1% increase in GDP is associated with a 4.6% increase in AI patents. Larger economies publish and patent more

Private investment is not a good predictor of patent output after controlling for GDP and region/year fixed effects

Predictor	Effect	Intepretation
Publications	Significant	Academic output corresponds to patent production
GDP	Significant	Larger economies produce more research and support more AI infrastructure
Private Investment	Not significant	Private investment may not have a direct pathway to patents

References

AI Index Steering Committee. (2025). *AI Index Report 2025*. Stanford Institute for Human-Centered Artificial Intelligence (HAI).

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World Bank. (2023). *World Development Indicators (WDI)* [Data set]. World Bank Group.

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Pavur, J., & Ding, J. (2023). *Patent Landscape for Computer Vision: United States and China*. Center for Security and Emerging Technology (CSET), Georgetown University.

<https://cset.georgetown.edu/publication/patent-landscape-for-computer-vision-united-states-and-china/>

Questions?
